



SEC 3212: Network Security

2nd Trimester: 2024

Network Security Course Project

Virtual Network Using VirtualBox with SSH Access

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Overview:

Secure Shell (SSH) is a network protocol that provides a secure method for accessing and managing networked devices over an unsecured network. It allows users to remotely log in to another computer and execute commands securely. SSH encrypts the communication between the client and the server, ensuring confidentiality and integrity of the transmitted data.

Key benefits of SSH:

- Secure communication over unsecured networks.
- Remote administration of servers and devices.
- File transfer using SCP or SFTP.
- Port forwarding and tunneling capabilities.

Tasks:

Complete the following tasks and provide screenshots for each step:

A. Verify Network Configuration

1. Verify the network configuration for each machine using ip addr.

```
(kali@KaliBase)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:17:57:5e brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 84416sec preferred_lft 84416sec
   inet6 fe80::f768:e080:fa8c:ae9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:99:50:d3 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.101/24 brd 192.168.56.255 scope global eth1
       valid_lft forever preferred_lft forever
   inet6 fe80::a00:27ff:fe99:50d3/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
(kali@KaliBase)-[~]
$
```

```

(kali㉿Kali-1)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f7:20:f0 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86271sec preferred_lft 86271sec
    inet6 fe80::d64d:5b01:125c:a5f6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:34:2d:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe34:2d02/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

```

(kali㉿Kali-2)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:47:bd:21 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86287sec preferred_lft 86287sec
    inet6 fe80::416b:c919:63b6:871a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5b:87:f7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe5b:87f7/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

The command **ip addr** was executed on kali base , kali-1 and kali-2 to display network information. Shows details about various network interfaces, such as **lo** (loopback interface) and **eth0** (network interface). The status "**UP**" is indicated in all, meaning the interfaces are active and functioning correctly, also show different IP addresses assigned to each interface, reflecting the network configuration in each system.

2. Ping ubuntu-1 and ubuntu-2 from Ubuntu Base.

```
(kali㉿KaliBase)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
 64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.35 ms
 64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.72 ms
 64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=8.01 ms
 64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.963 ms
 64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=2.63 ms
^C
— 192.168.56.102 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.963/2.934/8.008/2.596 ms

(kali㉿KaliBase)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
 64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=1.69 ms
 64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=2.16 ms
 64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.75 ms
 64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=1.68 ms
^C
— 192.168.56.103 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.681/1.821/2.159/0.196 ms

(kali㉿KaliBase)-[~]
$ █
```

We executed the **ping** command from KaliBase to kali-1 and kali-2 to test the communication between the systems. The results showed successful connectivity, with replies received and response times, indicating that the devices are connected and functioning properly.

B. Configure SSH on All Machines.

1. Establish SSH connections:

- Ubuntu Base to ubuntu-1.

```
(kali㉿KaliBase)-[~]
$ ssh kali@192.168.56.102
kali@192.168.56.102's password:
Linux Kali-1 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(kali㉿Kali-1)-[~]
$ █
```

SSH was configured on all systems to enable secure remote connections. The command **ssh kali@192.168.56.102** was used from the KaliBase to connect to kali-1, resulting in a successful login after entering the correct password. The system displayed a message indicating it runs on Kali Linux, confirming that the connection was established properly.

- Ubuntu Base to ubuntu-2.

```
(kali@KaliBase)-[~]
$ ssh kali@192.168.56.103
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:1keFUQeZoC0rBxkRCOhgXabeAryhURszg03rkzBu5J0.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.103' (ED25519) to the list of known hosts.
kali@192.168.56.103's password:
Linux Kali-2 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:05:52 2025 from 192.168.56.102
(kali@Kali-2)-[~]
$
```

The command **ssh kali@192.168.56.103** was used from KaliBase to kali-2, but the authenticity of the host could not be established initially. A warning prompted the user to confirm whether to continue connecting. After entering "yes," the host key was permanently added to the list of known hosts. Subsequently, the system displayed information about the Kali Linux environment, confirming that the connection was successfully established.

- ubuntu-1 to ubuntu-2.

```
(kali@Kali-1)-[~]
$ ssh kali@192.168.56.103
kali@192.168.56.103's password:
Linux Kali-2 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:44:20 2025 from 192.168.56.101
(kali@Kali-2)-[~]
$
```

The command **ssh kali@192.168.56.103** was used from the kali-1 to connect to kali-2, resulting a successful login.

2. Create a text file on ubuntu-1 from Ubuntu Base via SSH.

```
(kali@KaliBase)-[~/Desktop]
$ ssh kali@192.168.56.102
kali@192.168.56.102's password:
Linux Kali-1 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:30:46 2025 from 192.168.56.101
(kali@Kali-1)-[~]
$ echo "This is a test file from Kali Base." > testfile.txt

(kali@Kali-1)-[~]
$ ls
check_money_tiny      Desktop      _MACOSX      money.h       scanresults.html  upper.c
check_money_tiny.c    Documents   money.c       money.o       scan.xml          Videos
check_money_tiny.gcda Downloads   money.c.gcov  Music        student-copy      xml.scan
check_money_tiny.gcno hello.c     money.gcda    Pictures     Templates
check_money_tiny.o    'Lab 4.pdf' money.gcno     Public       testfile.txt
```

After establishing the connection, a text file **testfile.txt** was created with the content "This is a text file from Kali Base." To verify that the file was successfully created, we used the **ls** command.

3. Restart the SSH service on ubuntu-1 from Ubuntu Base.

```
(kali@Kali-1)-[~]
$ sudo systemctl restart ssh
[sudo] password for kali:

(kali@Kali-1)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-01-18 05:47:47 EST; 14s ago
     Invocation: 141912cae9fb4cedadccffde85a678ae
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 11945 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 11948 (sshd)
      Tasks: 1 (limit: 2220)
     Memory: 1.1M (peak: 1.6M)
        CPU: 63ms
    CGroup: /system.slice/ssh.service
            └─11948 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 18 05:47:47 Kali-1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Jan 18 05:47:47 Kali-1 sshd[11948]: Server listening on 0.0.0.0 port 22.
Jan 18 05:47:47 Kali-1 sshd[11948]: Server listening on :: port 22.
Jan 18 05:47:47 Kali-1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(kali@Kali-1)-[~]
$
```

The first command **sudo systemctl restart ssh** is used to restart the ssh service, while the second command **sudo systemctl status ssh** checks the status of the service. The status was enabled, active (running), indicating that the service is functioning correctly.

The full picture:

```
(kali㉿KaliBase)-[~/Desktop]
$ ssh kali@192.168.56.102
kali@192.168.56.102's password:
Linux Kali-1 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

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individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:30:46 2025 from 192.168.56.101
(kali㉿Kali-1)-[~]
$ echo "This is a test file from Kali Base." > testfile.txt

(kali㉿Kali-1)-[~]
$ ls
check_money_tiny      Desktop      __MACOSX      money.h        scanresults.html  upper.c
check_money_tiny.c    Documents   money.c        money.o        scan.xml          Videos
check_money_tiny.gcd  Downloads   money.c.gcov   Music          student-copy      xml.scan
check_money_tiny.gcno hello.c     money.gcda     Pictures       Templates
check_money_tiny.o    'Lab 4.pdf' money.gcno     Public         testfile.txt

(kali㉿Kali-1)-[~]
$ █

(kali㉿Kali-1)-[~]
$ sudo systemctl restart ssh
[sudo] password for kali:

(kali㉿Kali-1)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-01-18 05:47:47 EST; 14s ago
     Invocation: 141912cae9fb4eedadcffdfef85a678ae
       Docs: man:sshd(8)
            man:sshd_config(5)
    Process: 11945 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 11948 (sshd)
      Tasks: 1 (limit: 2220)
     Memory: 1.1M (peak: 1.6M)
        CPU: 63ms
    CGroup: /system.slice/ssh.service
            └─11948 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 18 05:47:47 Kali-1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 18 05:47:47 Kali-1 sshd[11948]: Server listening on 0.0.0.0 port 22.
Jan 18 05:47:47 Kali-1 sshd[11948]: Server listening on :: port 22.
Jan 18 05:47:47 Kali-1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(kali㉿Kali-1)-[~]
$ █
```

4. Transfer a file from Ubuntu Base to ubuntu-2 using SCP.

The transfer was completed successfully at 100%.

```
(kali@KaliBase)-[~/Desktop]
$ scp testfile.txt kali@192.168.56.103:/home/kali
kali@192.168.56.103's password:
testfile.txt                                100%   0   0.0KB/s   00:00

(kali@KaliBase)-[~/Desktop]
$ ssh kali@192.168.56.103
kali@192.168.56.103's password:
Linux Kali-2 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:36:33 2025 from 192.168.56.101
(kali@Kali-2)-[~]
$ ls
check_money_tiny      Desktop      __MACOSX      money.h        scanresults.html  upper.c
check_money_tiny.c    Documents   money.c        money.o        scan.xml          Videos
check_money_tiny.gcda Downloads    money.c.gcov   Music          student-copy      xml.scan
check_money_tiny.gcno 'Lab 4.pdf' money.gcda     Pictures       Templates
check_money_tiny.o   money.gcno  Public         testfile.txt

(kali@Kali-2)-[~]
$
```

C. Firewall Configuration Instructions

- Firewall rules (sudo ufw status).

```
(kali@KaliBase)-[~]
$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Anywhere ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Anywhere (v6) ALLOW Anywhere (v6)

(kali@Kali-1)-[~]
$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Anywhere ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Anywhere (v6) ALLOW Anywhere (v6)
```

```
(kali㉿Kali-2)-[~]
$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Anywhere ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Anywhere (v6) ALLOW Anywhere (v6)
```

The command **sudo ufw status** was used to check the firewall status. The results show in the pictures above that the firewalls are active and display some rules (ex: Allowing TCP connections on port 22 for ssh from anywhere).

- Successful SSH connections after the firewall is enabled.

Successful

```
(kali㉿KaliBase)-[~]
$ ssh kali@192.168.56.102
kali@192.168.56.102's password:
Linux Kali-1 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:45:50 2025 from 192.168.56.101
(kali㉿Kali-1)-[~]
$ ls
check_money_tiny Desktop __MACOSX money.h scanresults.html upper.c
check_money_tiny.c Documents money.c money.o scan.xml Videos
check_money_tiny.gcda Downloads money.c.gcov Music student-copy xml.scan
check_money_tiny.gcno hello.c money.gcda Pictures Templates
check_money_tiny.o 'Lab 4.pdf' money.gcn0 Public testfile.txt

(kali㉿Kali-1)-[~]
$ exit
Connection to 192.168.56.102 closed.

(kali㉿KaliBase)-[~]
$
```

Successful

```
(kali㉿KaliBase)-[~]
$ ssh kali@192.168.56.103
kali@192.168.56.103's password:
Linux Kali-2 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 05:50:38 2025 from 192.168.56.102
(kali㉿Kali-2)-[~]
$ ls
check_money_tiny      Desktop      __MACOSX      money.h      scanresults.html  upper.c
check_money_tiny.c    Documents   money.c        money.h      scan.xml          Videos
check_money_tiny.gcda Downloads    money.c.gcov   money.o      student-copy      xml.scan
check_money_tiny.gcno hello.c      money.gcda     Music        Templates
check_money_tiny.o    'Lab 4.pdf' money.gcno     Pictures     testfile.txt
Public

(kali㉿Kali-2)-[~]
$ exit
Connection to 192.168.56.103 closed.

(kali㉿KaliBase)-[~]
$ █
```

Successful

```
(kali㉿Kali-1)-[~]
$ ssh kali@192.168.56.103
kali@192.168.56.103's password:
Linux Kali-2 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 18 06:16:18 2025 from 192.168.56.101
(kali㉿Kali-2)-[~]
$ ls
check_money_tiny      Desktop      __MACOSX      money.h      scanresults.html  upper.c
check_money_tiny.c    Documents   money.c        money.h      scan.xml          Videos
check_money_tiny.gcda Downloads    money.c.gcov   money.o      student-copy      xml.scan
check_money_tiny.gcno hello.c      money.gcda     Music        Templates
check_money_tiny.o    'Lab 4.pdf' money.gcno     Pictures     testfile.txt
Public

(kali㉿Kali-2)-[~]
$ exit
Connection to 192.168.56.103 closed.

(kali㉿Kali-1)-[~]
$ █
```

- Ping results to verify ICMP traffic is working.

Ping is working

```
(kali㉿KaliBase)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.926 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.673 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.15 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.725 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.897 ms
^C
— 192.168.56.102 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4037ms
rtt min/avg/max/mdev = 0.673/1.075/2.154/0.548 ms

(kali㉿KaliBase)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=2.02 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.31 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=2.10 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.806 ms
^C
— 192.168.56.103 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.806/1.559/2.103/0.532 ms

(kali㉿KaliBase)-[~]
$ █
```

Ping is working

```
(kali㉿Kali-1)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.929 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.473 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.572 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.525 ms
^C
— 192.168.56.103 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3026ms
rtt min/avg/max/mdev = 0.473/0.624/0.929/0.179 ms

(kali㉿Kali-1)-[~]
$ █
```

D. Monitoring Network Traffic with Wireshark

1. Install Wireshark

1. Install Wireshark on the Ubuntu Base VM.

First, we will update the package list to ensure we have the latest information on available packages.

```
(ghadah@kali-base)-[~]
$ sudo apt update
[sudo] password for ghadah:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
920 packages can be upgraded. Run 'apt list --upgradable' to see them.
(ghadah@kali-base)-[~]
$
```

Command `sudo apt install wireshark` was used to install Wireshark.

```
l$ sudo apt install wireshark
Upgrading:
 libwireshark-data libwireshark18 libwiretap15 libwsutil16 tshark wireshark wireshark-common

Summary:
  Upgrading: 7, Installing: 0, Removing: 0, Not Upgrading: 913
  Download size: 27.3 MB
  Freed space: 435 kB

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 libwireshark-data all 4.4.3-1 [451 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libwsutil16 amd64 4.4.3-1 [128 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libwiretap15 amd64 4.4.3-1 [270 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libwireshark18 amd64 4.4.3-1 [21.1 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 tshark amd64 4.4.3-1 [177 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 wireshark-common amd64 4.4.3-1 [600 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 wireshark amd64 4.4.3-1 [4,619 kB]
Fetched 27.3 MB in 5s (5,888 kB/s)
Preconfiguring packages ...
(Reading database ... 4080 files and directories currently installed.)
Preparing to unpack .../0-libwireshark-data_4.4.3-1_all.deb ...
Unpacking libwireshark-data (4.4.3-1) over (4.4.0-1) ...
Preparing to unpack .../1-libwsutil16_4.4.3-1_amd64.deb ...
Unpacking libwsutil16:amd64 (4.4.3-1) over (4.4.0-1) ...
Preparing to unpack .../2-libwiretap15_4.4.3-1_amd64.deb ...
Unpacking libwiretap15:amd64 (4.4.3-1) over (4.4.0-1) ...
Preparing to unpack .../3-libwireshark18_4.4.3-1_amd64.deb ...
Unpacking libwireshark18:amd64 (4.4.3-1) over (4.4.0-1) ...
Preparing to unpack .../4-tshark_4.4.3-1_amd64.deb ...
Unpacking tshark (4.4.3-1) over (4.4.0-1) ...
Preparing to unpack .../5-wireshark-common_4.4.3-1_amd64.deb ...
Unpacking wireshark-common (4.4.3-1) over (4.4.0-1) ...
Preparing to unpack .../6-wireshark_4.4.3-1_amd64.deb ...
Unpacking wireshark (4.4.3-1) over (4.4.0-1) ...
Setting up libwireshark-data (4.4.3-1) ...
Setting up libwsutil16:amd64 (4.4.3-1) ...
Setting up libwiretap15:amd64 (4.4.3-1) ...
Setting up libwireshark18:amd64 (4.4.3-1) ...
Setting up wireshark-common (4.4.3-1) ...
Setting up wireshark (4.4.3-1) ...
Setting up tshark (4.4.3-1) ...
Processing triggers for kali-menu (2024.4.0) ...
Processing triggers for desktop-file-utils (0.27-2) ...
Processing triggers for hicolor-icon-theme (0.18-1) ...
Processing triggers for libc-bin (2.40-3) ...
Processing triggers for man-db (2.13.0-1) ...
Processing triggers for shared-mime-info (2.4-5+b1) ...

Progress: [ 97%] |
```

2. During installation, allow non-root users to capture packets if prompted.
The Wireshark we installed allow non-root users to capture packets by default

2. Capture Network Traffic

1. Open Wireshark on Ubuntu Base and observe the traffic.

By typing Wireshark in the terminal, Wireshark will be opened.

```
(ghadah@kali-base)-[~]
$ wireshark
** (wireshark:76587) 00:27:57.927053 [Capture MESSAGE] -- Capture Start ...
** (wireshark:76587) 00:27:58.028230 [Capture MESSAGE] -- Capture started
** (wireshark:76587) 00:27:58.029824 [Capture MESSAGE] -- File: "/tmp/wireshark_eth1GVVS02.pcapng"
** (wireshark:76587) 00:32:43.908021 [Capture MESSAGE] -- Capture Stop ...
```

3. Generate Network Traffic and Analyze it.

- Identify the IP addresses involved in the communication.
- Capture screenshots of the traffic logs and analyze the following:
 - Source and destination IP addresses.
 - Protocol types (SSH, ICMP, TCP).
 - Packet size and timing.

1. From Ubuntu Base, SSH into ubuntu-1 and ubuntu-2.

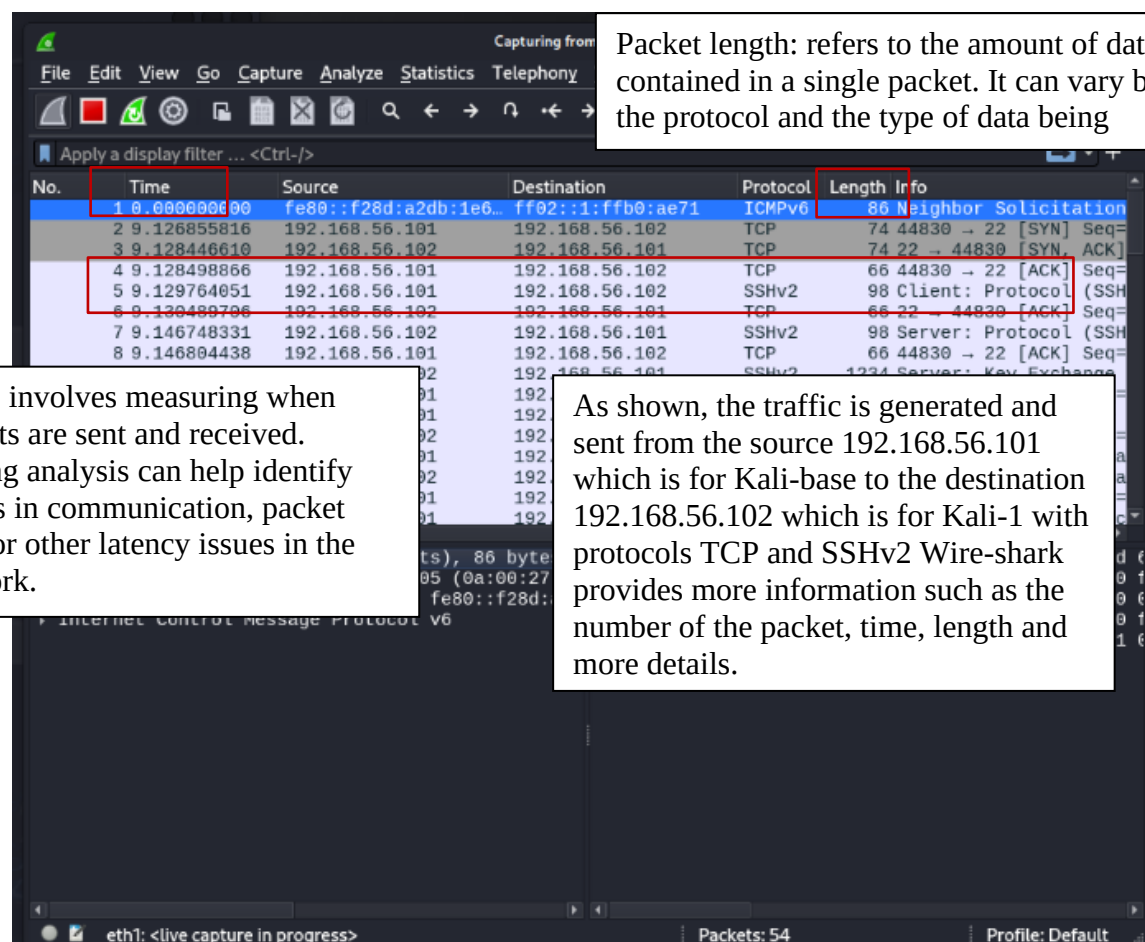
By typing command **ssh username@ ip address for kali-1** as shown we will enter from kali-base to kali-1 and we will generate traffic with ssh protocol in the Wireshark

```
(ghadah@kali-base)-[~]
$ ssh ghadah@192.168.56.102
ghadah@192.168.56.102's password:
Linux kali-1 6.11.2-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
last login: Sat Jan 18 12:37:00 2025 from 192.168.56.101
(ghadah@kali-1)-[~]
$
```

Here is the generated traffic for ssh connection from kali-base to kali-1

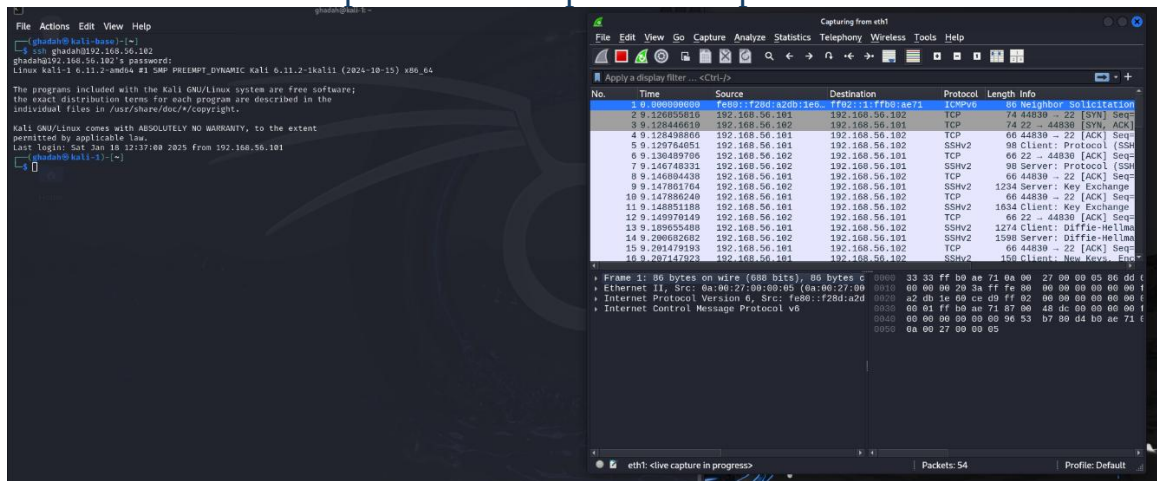


Packet length: refers to the amount of data contained in a single packet. It can vary based on the protocol and the type of data being

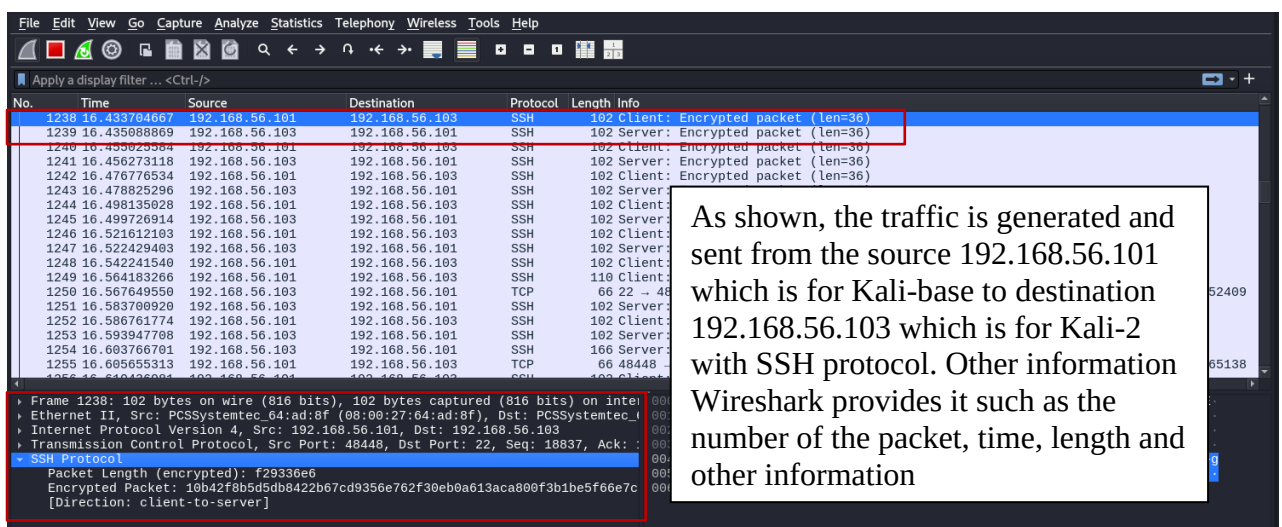
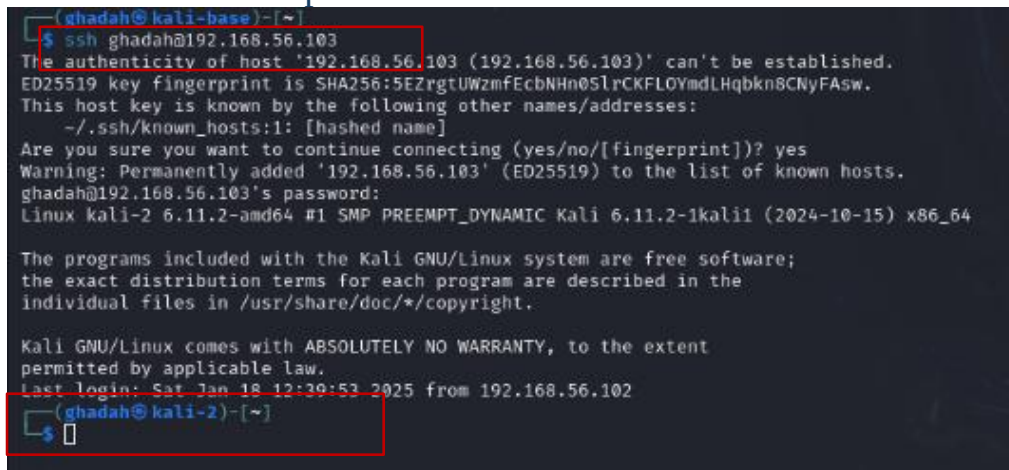
Time: involves measuring when packets are sent and received. Timing analysis can help identify delays in communication, packet loss, or other latency issues in the network.

As shown, the traffic is generated and sent from the source 192.168.56.101 which is for Kali-base to the destination 192.168.56.102 which is for Kali-1 with protocols TCP and SSHv2 Wire-shark provides more information such as the number of the packet, time, length and more details.

And here is the whole picture for the previous explanation:

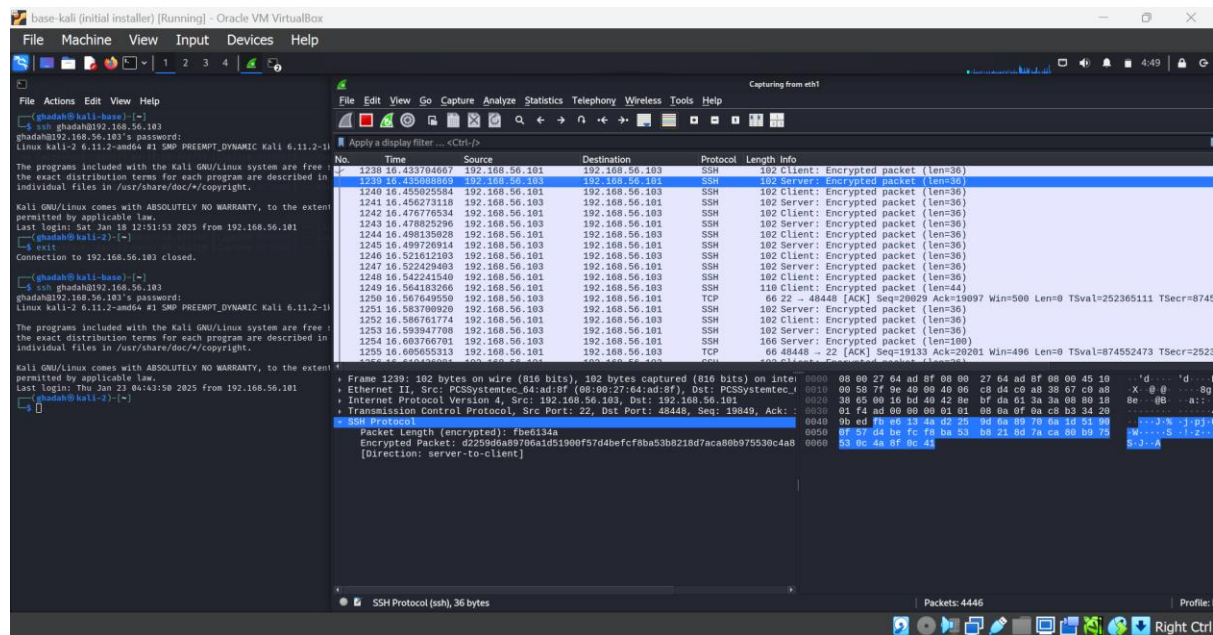


Also, here by typing command **ssh username@ ip address** for kali-2 as shown we will enter from kali-base to kali-2 and the traffic will be generated in Wireshark with ssh protocol



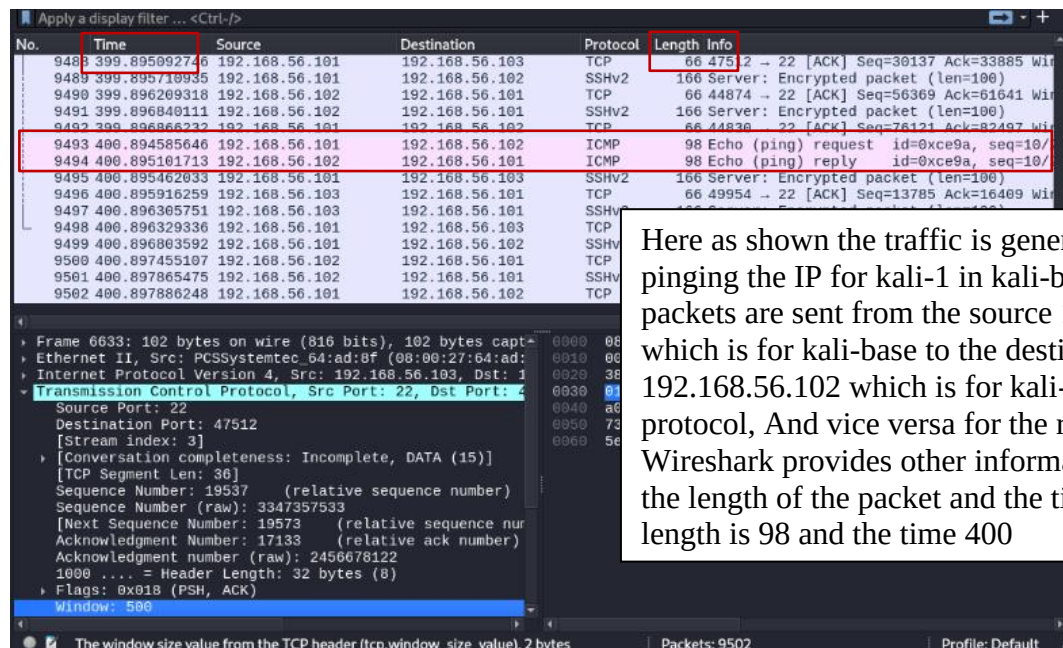
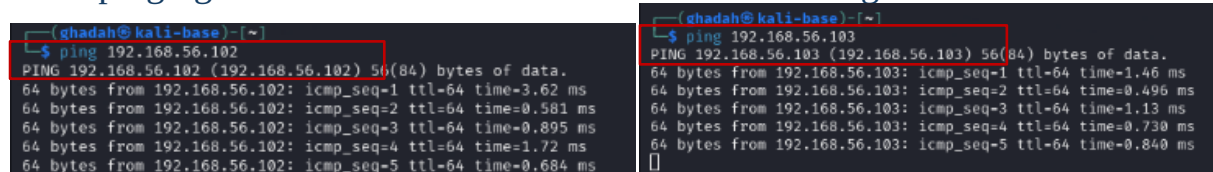
Also, here we have a lot of information for specific packet, just clicking on it will shows: the encrypted packet and the direction for the packet, which is here from client to server, and more.

And here is the whole picture for the previous explanation:

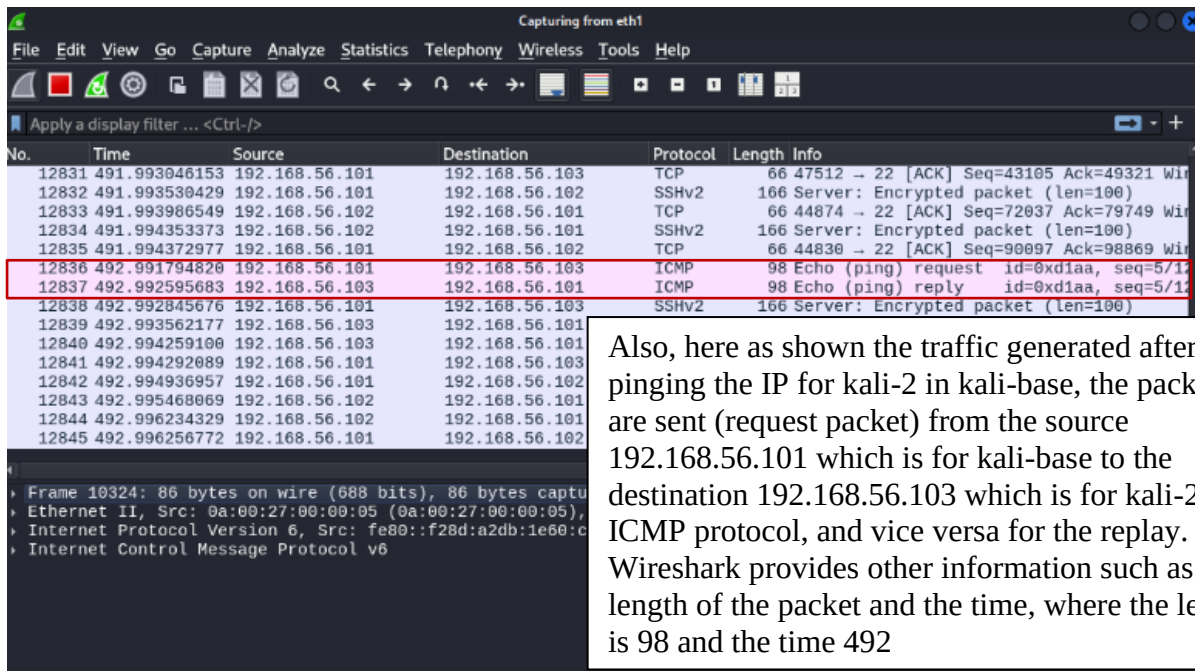


2. Ping ubuntu-1 and ubuntu-2 to generate ICMP traffic.

After pinging the two machines the ICMP traffic will be generated in Wireshark

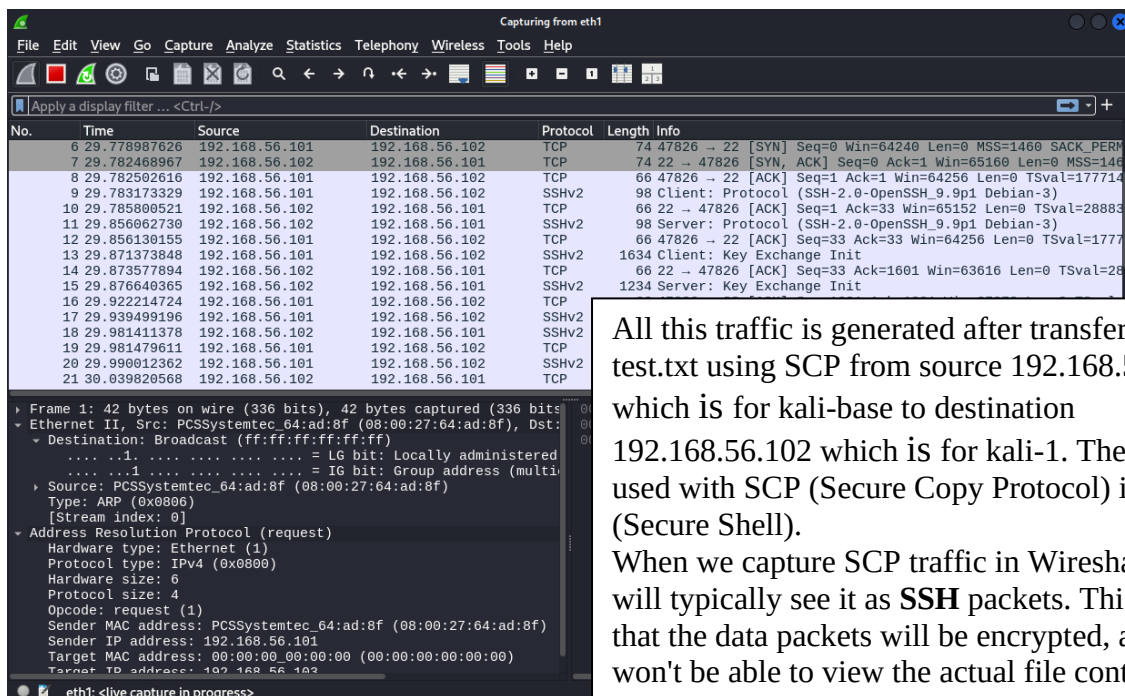
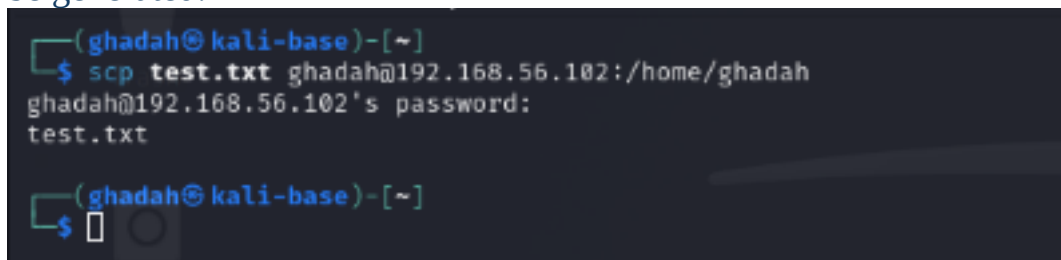


Here as shown the traffic is generated after pinging the IP for kali-1 in kali-base, the packets are sent from the source 192.168.56.101 which is for kali-base to the destination 192.168.56.102 which is for kali-1 with ICMP protocol, And vice versa for the replay. Wireshark provides other information such as the length of the packet and the time where the length is 98 and the time 400



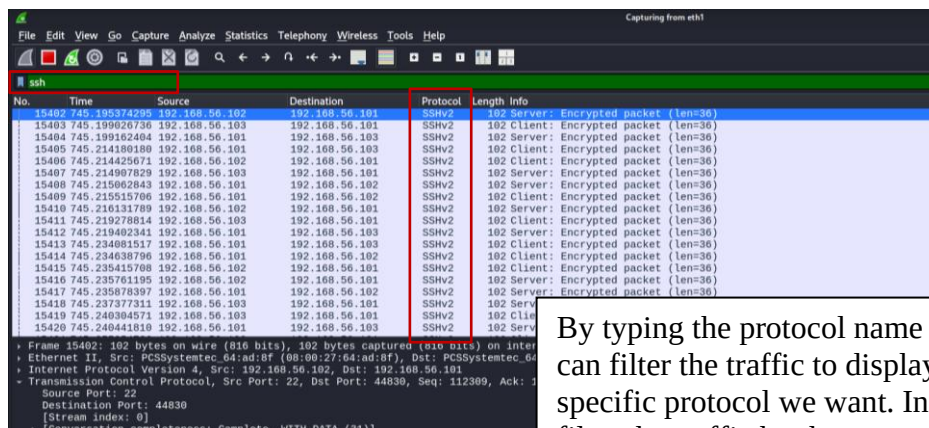
3. Use SCP to transfer a file between machines.

By typing command `scp fileName userName@ kali-1 ip` then the traffic will be generated.



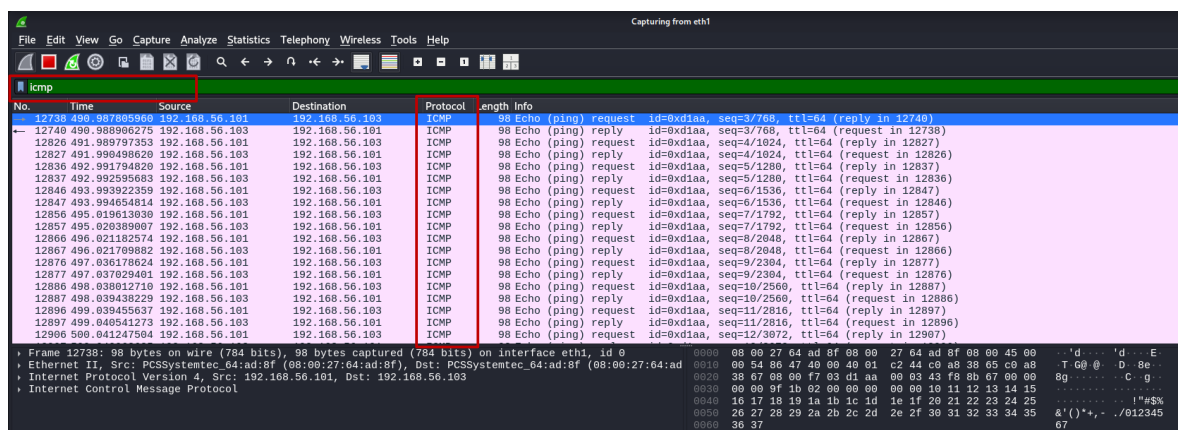
4. Filter traffic by protocol, such as:

- ssh to analyze SSH traffic.

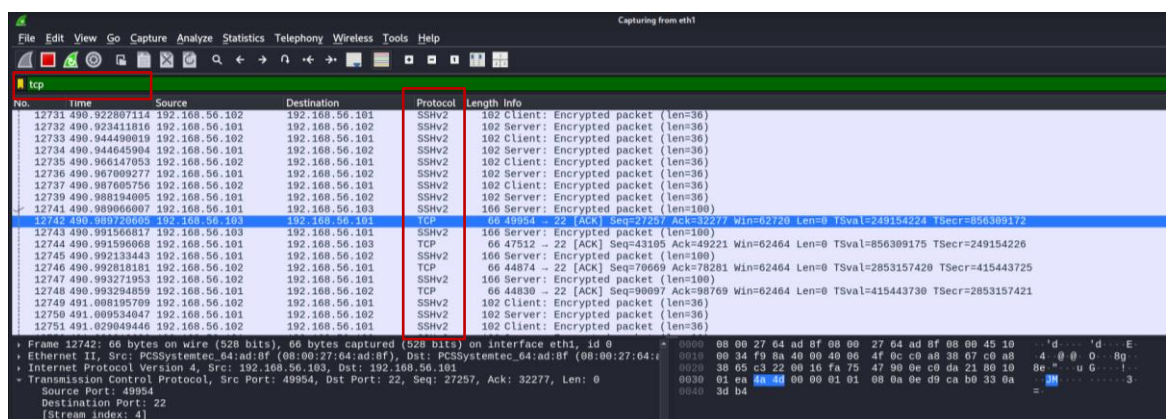


By typing the protocol name in the search bar, we can filter the traffic to display the packet with the specific protocol we want. In this example, we filter the traffic by the protocol name, SSH, ICMP and TCP

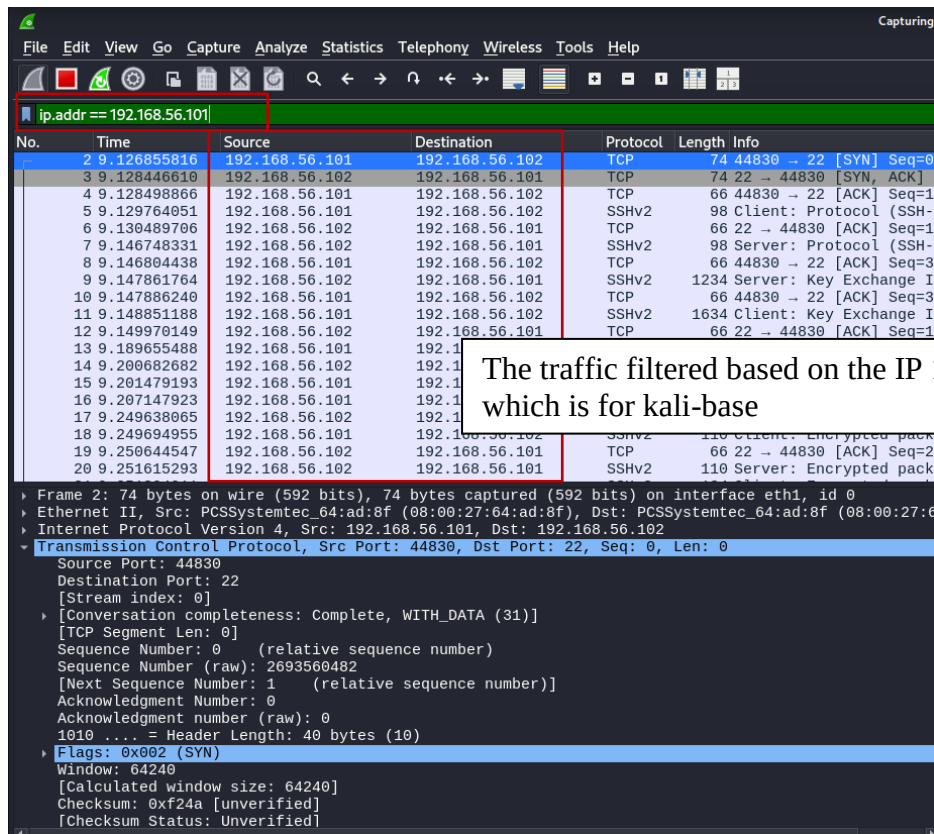
- icmp to view ping (ICMP) packets.



- tcp to observe general TCP connections.



- Also, we can filter the traffic by specific IP address, for example:



The traffic filtered based on the IP 192.168.56.101 which is for kali-base

No.	Time	Source	Destination	Protocol	Length	Info
2	9.126855816	192.168.56.101	192.168.56.102	TCP	74	44830 → 22 [SYN] Seq=0
3	9.128446610	192.168.56.102	192.168.56.101	TCP	74	22 → 44830 [SYN, ACK] Seq=1
4	9.128498866	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=1
5	9.129764051	192.168.56.101	192.168.56.102	SSHv2	98	Client: Protocol (SSH-)
6	9.130489706	192.168.56.102	192.168.56.101	TCP	66	22 → 44830 [ACK] Seq=1
7	9.146748331	192.168.56.102	192.168.56.101	SSHv2	98	Server: Protocol (SSH-)
8	9.146804438	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
9	9.147861764	192.168.56.102	192.168.56.101	SSHv2	1234	Server: Key Exchange I
10	9.147886240	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
11	9.148851188	192.168.56.101	192.168.56.102	SSHv2	1634	Client: Key Exchange I
12	9.149970149	192.168.56.102	192.168.56.101	TCP	66	22 → 44830 [ACK] Seq=1
13	9.189655488	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
14	9.200682682	192.168.56.102	192.168.56.101	TCP	66	22 → 44830 [ACK] Seq=1
15	9.201479193	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
16	9.207147923	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
17	9.249638065	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
18	9.249694955	192.168.56.101	192.168.56.102	TCP	66	44830 → 22 [ACK] Seq=3
19	9.250644547	192.168.56.102	192.168.56.101	TCP	66	22 → 44830 [ACK] Seq=1
20	9.251615293	192.168.56.102	192.168.56.101	SSHv2	110	Server: Encrypted pack

Frame 2: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth1, id 0
 Ethernet II, Src: PCSSystemtec_64:ad:8f (08:00:27:64:ad:8f), Dst: PCSSystemtec_64:ad:8f (08:00:27:64:ad:8f)
 Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.102
 Transmission Control Protocol, Src Port: 44830, Dst Port: 22, Seq: 0, Len: 0
 Source Port: 44830
 Destination Port: 22
 [Stream index: 0]
 [Conversation completeness: Complete, WITH_DATA (31)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 2693560482
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 0
 Acknowledgment number (raw): 0
 1010 = Header Length: 40 bytes (10)
 Flags: 0x002 (SYN)
 Window: 64240
 [Calculated window size: 64240]
 Checksum: 0xf24a [unverified]
 [Checksum Status: Unverified]